

1. Write a Python program to plot a few activation functions that are being used in neural networks.

2. Generate ANDNOT function using McCulloch-Pitts neural net by a python program.

3. Write a Python Program using Perceptron Neural Network to recognize even and odd numbers. Given numbers are in ASCII form 0 to 9

4. With a suitable example demonstrate the perceptron learning law with its decision regions using python. Give the output in graphical form.

5.. Write a python Program for Bidirectional Associative Memory with two pairs of vectors

6. Create a Neural network architecture from scratch in Python and use it to do multi-class classification on any data.

Parameters to be considered while creating the neural network from scratch are specified as:

(1) No of hidden layers : 1 or more

(2) No. of neurons in hidden layer: 100

(3) Non-linearity in the layer : Relu

(4) Use more than 1 neuron in the output layer. Use a suitable threshold value

Use appropriate Optimisation algorithm

7. Implement Artificial Neural Network training process in Python by using Forward Propagation, Back Propagation

8. Write a python program to show Back Propagation Network for XOR function with Binary Input and Output

9. Write a python program in python program for creating a Back Propagation Feed-forward neural Network

10. Write a python program to design a Hopfield Netwrk which stores 4 vectors

11. Write Python program to implement CNN object detection. Discuss numerous performance evaluation metrics for evaluating the object detecting algorithms' performance.

12. How to Train a Neural Network with TensorFlow/Pytorch and evaluation of logistic regression using TensorFlow

13. TensorFlow/Pytorch implementation of CNN

14. MNIST Handwritten Character Detection using PyTorch, Keras and Tensorflow